

Elementary Problems

Circle

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Basic Work

1. Let $ABCD$ be a trapezoid with $AD \parallel BC$. Prove that $ABCD$ is cyclic if and only if $ABCD$ is an isosceles trapezoid.
2. Let A and B be two points on a circle Γ . Points C and D lie on the same side of the line AB such that C lies inside Γ and D lies outside Γ . Prove that $\angle ACB > \angle ADB$.
3. Two circles Γ_1 and Γ_2 intersect at A and B . A line through A meets Γ_1 and Γ_2 at C and D respectively such that A lies between C and D . A line through B meets Γ_1 and Γ_2 at E and F respectively such that B lies between E and F . Prove that $CE \parallel DF$.
4. Let C and D be two points on a chord AB of a circle Γ_1 such that C lies between A and D . A circle Γ_2 passing through C and D meets Γ_1 at E and F such that A, E, F, B lie in that order on Γ_1 . Prove that $\angle CEA = \angle BFD$.
5. (All-Russian MO 1964) A circle with centre O is inscribed in quadrilateral $ABCD$. Prove that $\angle AOB + \angle COD = 180^\circ$.
6. Let $ABCD$ be a cyclic quadrilateral lying on Γ such that $AC \perp BD$. The tangents at A, B, C, D to Γ form a quadrilateral $WXYZ$. Prove that $WXYZ$ is cyclic.
7. Let B and D be two points on the circle with diameter AC and centre O such that B and D lie on opposite sides of the line AC . Suppose $\angle BAC = \angle COD$. Let E be the midpoint of the minor arc BC . Prove that $AC \perp DE$.
8. Let Γ be the circumcircle of $\triangle ABC$. Let M be a point on the arc AB of Γ not containing C such that $MA = MC$, and let N be a point of the arc AC of Γ not containing B such that $NA = NB$. Prove that MN is perpendicular to the internal angle bisector of $\angle BAC$.
9. Two circles Γ_1 and Γ_2 with centres O_1 and O_2 respectively intersect at A and B . The tangent at A to Γ_1 meets BO_2 at D . The tangent at A to Γ_2 meets BO_1 at C . Prove that A, C, B, D are concyclic, and prove that $CD \parallel O_1O_2$.
10. (Iranian Geometry Olympiad 2016 Medium) Let ω be the circumcircle of right-angled triangle ABC ($\angle A = 90^\circ$). Tangent to ω at point A intersects the line BC in point P . Suppose that M is the midpoint of (the smaller) arc AB , and PM intersects ω for the second time in Q . Tangent to ω at point Q intersects AC in K . Prove that $\angle PKC = 90^\circ$.
11. (IMO shortlist 1966) Show that the four perpendiculars dropped from the midpoints of the sides of a cyclic quadrilateral to the respective opposite sides are concurrent.

Power

12. Two circles Γ_1 and Γ_2 are internally tangent at X . Let P be an arbitrary point on the common tangent at X . A line ℓ through P meets Γ_1 at A and B . A line through P distinct from ℓ meets Γ_2 at C and D . Prove that A, B, C, D are concyclic.
13. Let M and N be the midpoints of the sides AB and AC of $\triangle ABC$ respectively. The circumcircle of $\triangle BCN$ meets AB at D . Prove that AC is a tangent to the circumcircle of $\triangle MND$.
14. Let H be a point on the altitude AD of an acute $\triangle ABC$. If $DB \cdot DC = DA \cdot DH$, prove that H is the orthocentre of $\triangle ABC$.
15. Let D and E be the feet of the altitudes from B and C of $\triangle ABC$ respectively. Let H be the orthocentre of $\triangle ABC$. A circle passing through A and E is tangent to the side BC at P . Prove that $\angle CPD = \angle PHD$.
16. (Iran MO 2nd Round 2004 Problem 5) The interior bisector of $\angle A$ from $\triangle ABC$ intersects the side BC and the circumcircle of $\triangle ABC$ at D, M , respectively. Let ω be a circle with centre M and radius MB . A line passing through D , intersects ω at X, Y . Prove that AD bisects $\angle XAY$.
17. (USAMO 1990 Problem 5) An acute-angled triangle ABC is given in the plane. The circle with diameter AB intersects altitude CC' and its extension at points M and N , and the circle with diameter AC intersects altitude BB' and its extensions at P and Q . Prove that the points M, N, P, Q lie on a common circle.
18. Let A, B, C, D be collinear points in that order. A circle Γ with centre O passes through B and C . Prove that $OA = OD$ if and only if $AB = CD$.
19. (All-Soviet Union MO 1973 Problem 4) Let OA and OB be tangents to a circle at A and B . The line parallel to OB through A meets the circle again at C . Suppose that the line OC meets the circle again at E , and the ray AE meets the line OB at K . Prove that K is the midpoint of OB .
20. Given an acute $\triangle ABC$, let D and E be the feet of the altitudes from B and C respectively. Let P be a point on the segment BD such that $\angle CPA = 90^\circ$. Let Q be a point on the segment CE such that $\angle AQB = 90^\circ$. Prove that $AP = AQ$.
21. Let D be a point on the side BC of $\triangle ABC$. Let M be the midpoint of BD and N the midpoint of CD . Denote by Γ the circumcircle of $\triangle AMC$. The line AD meets Γ again at E . The line EN meets Γ again at F . Prove that CF is tangent to the circumcircle of $\triangle ABC$.

22. In $\triangle ABC$, the internal angle bisector of $\angle BAC$ meets BC and the circumcircle of $\triangle ABC$ at D and E respectively. Let X and Y be the contact points of the tangents from E to the circle with diameter AD . Prove that B, C, X, Y are concyclic.
23. (Switzerland MO Final Round 2010) Let $\triangle ABC$ be a triangle with $AB \neq AC$. The incircle with centre I touches BC, CA, AB at D, E, F , respectively. Furthermore let M be the midpoint of EF and AD intersect the incircle at $P \neq D$.
Show that $PMID$ is cyclic.
24. In $\triangle ABC$, I is the incentre. The line through I perpendicular to AI meets the extension of BC at D . Let E be the foot of the perpendicular from I to AD . Prove that A, B, C, E are concyclic.
25. Let AB be a diameter of a circle Γ . Let C be an arbitrary point on Γ distinct from A and B . Let X and Y be the projection of A and B on the tangent at C respectively. Prove that $4AX \cdot BY = XY^2$.
26. (Romania District Olympiad 2006 Grade 7 Problem 4) Let ABC be a triangle with $AB = AC$. Let D be the midpoint of BC , M the midpoint of AD and N the foot of the perpendicular from D to BM . Prove that $\angle ANC = 90^\circ$.
27. (Japan MO Finals 2011 Problem 1) Given an acute triangle ABC with the midpoint M of BC , draw the perpendicular HP from the orthocentre H of $\triangle ABC$ to AM . Show that $AM \cdot PM = BM^2$.
28. In $\triangle ABC$, let D be the foot of the internal angle bisector of $\angle BAC$. Let M be the midpoint of AD . The perpendicular bisector of AD meets the line through D perpendicular to BC at E . Prove that E, M, B, C are concyclic.
29. The diagonals AC and BD of a convex quadrilateral $ABCD$ intersect at P . Let Q be the point on the extension of BA such that QC bisects $\angle DCA$. Suppose

$$PA \cdot PC + PA \cdot CD = PB \cdot PD.$$

Prove that B, C, D, Q are concyclic.

30. (Baltic Way 2006) In a triangle ABC , points D, E lie on sides AB, AC respectively. The lines BE and CD intersect at F . Prove that if

$$BC^2 = BD \cdot BA + CE \cdot CA,$$

then the points A, D, F, E lie on a circle.

31. In $\triangle ABC$, D is the midpoint of AC . The circumcircle of $\triangle ABD$ meets the side BC at E . The circumcircle of $\triangle ACE$ meets the extension of BA at F . The circle that passes through B and D and is tangent to BC at B meets the extension of BA at P . Prove that $AB = FP$.

32. Let I be the incentre of $\triangle ABC$. Let M and N be the midpoints of the arc BC of the circumcircle of $\triangle ABC$ such that N and A lie on the same side of BC . Points P and Q lie on the side BC such that P lies inside $\triangle ABM$ and Q lies inside $\triangle AMC$. A circle Γ_1 passing through B and P has its centre lying on the line BN . A circle Γ_2 passing through C and Q has its centre lying on the line CN . Suppose MP meets Γ_1 again at X , MQ meets Γ_2 again at Y . Prove that X, I, Y are collinear if and only if P, M, Q, I are concyclic.
33. (IMO 1983 Problem 2) Let A be one of the two distinct points of intersection of two unequal coplanar circles C_1 and C_2 with centres O_1 and O_2 , respectively. One of the common tangents to the circles touches C_1 at P_1 and C_2 at P_2 , while the other touches C_1 at Q_1 and C_2 at Q_2 . Let M_1 be the midpoint of P_1Q_1 , and M_2 the midpoint of P_2Q_2 . Prove that $\angle O_1AO_2 = \angle M_1AM_2$.
34. (IMO 2009 Problem 2) Let ABC be a triangle with circumcentre O . The points P and Q are interior points of the sides CA and AB , respectively. Let K, L and M be the midpoints of the segments BP, CQ and PQ , respectively, and let Γ be the circle passing through K, L and M . Suppose that the line PQ is tangent to the circle Γ . Prove that $OP = OQ$.

Radical Axis

35. Two circles Γ_1 and Γ_2 meet at points A and B . An external common tangent to Γ_1 and Γ_2 meets the two circles at C and D respectively. Prove that AB bisects CD .
36. Let D and E be points on the sides AB and AC of $\triangle ABC$ respectively such that $DA = DC$ and $EA = EB$. The circumcircles of $\triangle ABC$ and $\triangle ADE$ meet again at P . The line DE meets the line BC at Q . Prove that A, P, Q are collinear.
37. (IMO 1959 Problem 5) An arbitrary point M is selected in the interior of the segment AB . The square $AMCD$ and $MBEF$ are constructed on the same side of AB , with segments AM and MB as their respective bases. The circles circumscribed about these squares, with centres P and Q , intersect at M and also at another point N .
- (b) Prove that the straight lines MN pass through a fixed point S independent of the choice of M .
38. (IMO shortlist 1970) In the triangle ABC let B' and C' be the midpoints of the sides AC and AB respectively and H the foot of the altitude passing through the vertex A . Prove that the circumcircles of the triangles $AB'C'$, $BC'H$, and $B'CH$ have a common point I and that the line HI passes through the midpoint of the segment $B'C'$.
39. Let D and E be the midpoints of the sides AB and AC of $\triangle ABC$ respectively. Prove that the A -altitude is the radical axis of the circles with diameters CD and BE .

40. Let $ABCD$ is a trapezoid with $AD \parallel BC$. Let AB meet CD at E . The circle with diameter AC meets the circle with diameter BD at X and Y . Prove that E, X, Y are collinear.
41. Let D and E be points on the sides AB and AC of $\triangle ABC$ respectively such that $DE \parallel BC$. Let P be any point inside $\triangle ADE$. The line DE meets BP and CP at S and T respectively. The circumcircles of $\triangle PDT$ and $\triangle PES$ meet again at Q . Prove that A, P, Q are collinear.
42. Let D and E be the midpoints of the sides AB and AC of $\triangle ABC$ respectively. Let Γ be the circumcircle of $\triangle ABC$. The line CD meets Γ again at P . The line PE meets Γ again at Q . The circumcircles of $\triangle APD$ and $\triangle AEQ$ meet again at X . Prove that BQ, CP, AX are concurrent.
43. Let $ABCD$ be a convex quadrilateral lying inside a circle Γ . The line AB meets Γ at points W_1 and W_2 . The line CD meets Γ at points X_1 and X_2 . The line AD meets Γ at points Y_1 and Y_2 . The line BC meets Γ at points Z_1 and Z_2 . Suppose points A, B, X_1, X_2 lie on a circle Γ_1 , points B, C, Y_1, Y_2 lie on a circle Γ_2 , points C, D, W_1, W_2 lie on a circle Γ_3 .
- (a) Prove that points A, D, Z_1, Z_2 lie on a circle Γ_4 .
- (b) Prove that the centres of $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ form a parallelogram.
44. The incircle of $\triangle ABC$ touches the sides BC, CA, AB at D, E, F respectively. Let $\omega, \omega_1, \omega_2, \omega_3$ be the circumcircles of $\triangle ABC, \triangle AFE, \triangle BDF, \triangle CED$ respectively. Let ω and ω_1 intersect at A and X , ω and ω_2 intersect at B and Y , ω and ω_3 intersect at C and Z .
- (a) Prove that X, Y, D, E are concyclic.
- (b) Prove that DX, EY, FZ are concurrent.
45. (Turkey MO 2nd Round 2006 Problem 1) Points P and Q on side AB of a convex quadrilateral $ABCD$ are given such that $AP = BQ$. The circumcircles of triangles APD and BQD meet again at K and those of APC and BQC meet again at L . Show that the points D, C, K, L lie on a circle.
- (Remark: The condition $AP = BQ$ is not necessary.)
46. Let D, E, F be the contact points of the incircle to the sides BC, CA, AB of $\triangle ABC$ respectively. Let ℓ_b be the line equidistant to B and DF , and let ℓ_c be the line equidistant to C and DE . Suppose ℓ_b meets ℓ_c at P . Prove that $PB = PC$.
47. (Iran TST 2011 Problem 1) In acute triangle ABC , $\angle B$ is greater than $\angle C$. Let M be the midpoint of BC . D and E are the feet of the altitudes from C and B respectively. K and L are midpoints of ME and MD respectively. If KL intersects the line through A parallel to BC in T , prove that $TA = TM$.

48. Let D and E be the feet of the altitudes from B and C of an acute $\triangle ABC$ respectively. Let H be the orthocentre of $\triangle ABC$, and let P be the midpoint of AH . The line through A parallel to BC meets the line DE at Q . Prove that PQ is perpendicular to the median from A .
49. (IMO shortlist 1997) The altitudes through the vertices A, B, C of an acute-angled triangle ABC meet the opposite sides at D, E, F , respectively. The line through D parallel to EF meets the lines AC and AB at Q and R , respectively. The line EF meets BC at P . Prove that the circumcircle of the triangle PQR passes through the midpoint of BC .

Record

v1.0: problems 1–49